Numerical Integration

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Lesson 3

1. Introduction to Numerical Integration

Integration is an important mathematical operation used to determine quantities such as area, volume, distance, accumulated change, and average values.

A definite integral is written as

$$I = \int_a^b f(x)\,dx$$

where:

- a is the lower limit of integration,
- b is the upper limit of integration,
- $f(x)$ is the integrand,
- $[a, b]$ is the interval of integration.

If the antiderivative of $f(x)$ can be found easily, the integral can be evaluated analytically.

For example,

$$\int_0^2 x^2\,dx$$

can be evaluated as

$$\int_0^2 x^2\,dx = \left[\frac{x^3}{3}\right]_0^2$$

Therefore,

$$\int_0^2 x^2\,dx = \frac{2^3}{3} - \frac{0^3}{3}$$

$$= \frac{8}{3}$$

Hence,

$$\boxed{\int_0^2 x^2\,dx = \frac{8}{3}}$$

However, in many practical situations, an exact analytical solution may not be available.

For example,

$$\int_0^1 e^{-x^2}\,dx$$

does not have an elementary antiderivative. Therefore, numerical methods can be used to obtain an approximate value.

- **Numerical integration** provides methods for approximating the value of a definite integral using a finite number of function values.

- The main idea is to replace the original function by a simpler function whose integral can be calculated easily.

2. Why Numerical Integration is Needed

Numerical integration is used when:

- ❑The antiderivative of a function cannot be obtained easily.
- ❑The function is available only at discrete data points.
- ❑The function is obtained from experimental observations.
- ❑The analytical integration is complicated.
- ❑An approximate numerical value is sufficient.

For example, suppose the following data are available:

x	0	1	2	3	4
$f(x)$	2.1	3.5	5.2	7.8	10.4

In this case, we may not know the exact mathematical function $f(x)$. However, numerical integration methods can be used to approximate

$$\int_0^4 f(x)\,dx.$$

3. Definite Integrals

Consider the definite integral

$$I = \int_a^b f(x)\,dx.$$

If $f(x) \geq 0$ over the interval $[a, b]$, the definite integral represents the area under the curve

$$y = f(x)$$

between

$$x = a$$

and

$$x = b.$$

For example,

$$I = \int_0^2 x^2 \, dx$$

represents the area under the curve

$$y = x^2$$

between $x = 0$ and $x = 2$.

4. Newton–Cotes Integration Formulas

- Newton–Cotes formulas are numerical integration formulas obtained by approximating a function using interpolation polynomials at equally spaced points.

The most commonly used Newton–Cotes formulas are:

- Trapezoidal Rule
- Composite Trapezoidal Rule
- Simpson's 1/3 Rule
- Composite Simpson's 1/3 Rule
- Simpson's 3/8 Rule

5.1 Trapezoidal Rule

Consider the definite integral

$$I = \int_{x_0}^{x_1} f(x)\,dx.$$

Let

$$x_1 = x_0 + h,$$

where h is the step size.

The function $f(x)$ is approximated by the straight line joining the points

$$(x_0, f(x_0))$$

and

$$(x_1, f(x_1)).$$

The area under this straight-line approximation is a trapezoid.

The area of a trapezoid is

$$A = \frac{1}{2}(\text{sum of parallel sides})(\text{height}).$$

Therefore,

$$\int_{x_0}^{x_1} f(x)\,dx \approx \frac{x_1 - x_0}{2}\left[f(x_0) + f(x_1)\right].$$

Since

$$x_1 - x_0 = h,$$

we obtain

$$\boxed{\int_{x_0}^{x_1} f(x)\,dx \approx \frac{h}{2}\left[f(x_0) + f(x_1)\right]}$$

This is called the **Trapezoidal Rule.**

Error term

The error in the Trapezoidal Rule is

$$\boxed{E_T = -\frac{h^3}{12} f''(\xi)}$$

for some

$$\xi \in (x_0, x_1).$$

Therefore, the exact integral can be written as

$$\boxed{\int_{x_0}^{x_1} f(x)\,dx = \frac{h}{2}\left[f(x_0) + f(x_1)\right] - \frac{h^3}{12} f''(\xi)}$$

for some $\xi \in (x_0, x_1)$.

Hence, the order of the error is

$$\boxed{E_T = O(h^3)}$$

Example 1

Use the **Trapezoidal Rule** to approximate

$$I = \int_0^2 x^2 \, dx$$

using one subinterval.

The Trapezoidal Rule is

$$\int_{x_0}^{x_1} f(x)\,dx \approx \frac{h}{2}\left[f(x_0) + f(x_1)\right]$$

Here,

$$a = 0, \qquad b = 2.$$

Since we are using one subinterval,

$$x_0 = 0, \qquad x_1 = 2.$$

The step size is

$$h = x_1 - x_0 = 2.$$

The function is

$$f(x) = x^2.$$

Therefore,

$$f(0) = 0^2 = 0$$

and

$$f(2) = 2^2 = 4.$$

Substitute these values into the Trapezoidal Rule:

$$I_T = \frac{2}{2}\left[f(0) + f(2)\right].$$

Thus,

$$I_T = 1(0 + 4).$$

Therefore,

$$\boxed{I_T = 4}$$

Exact Value

The exact value is

$$I = \int_0^2 x^2 \, dx.$$

Integrating,

$$I = \left[\frac{x^3}{3}\right]_0^2.$$

Therefore,

$$I = \frac{2^3}{3} - \frac{0^3}{3}$$

$$I = \frac{8}{3}.$$

Hence,

$$\boxed{I = 2.6667}$$

Absolute Error

The absolute error is

$$E_a = |I - I_T|.$$

Therefore,

$$E_a = |2.6667 - 4|.$$

Hence,

$$\boxed{E_a \approx 1.3333}$$

Example 2

The following values of a function are given:

x	1	2
$f(x)$	3	7

Use the Trapezoidal Rule to approximate

$$\int_1^2 f(x)\,dx.$$

Solution

The formula is

$$I_T = \frac{h}{2}\left[f(x_0) + f(x_1)\right].$$

Here,

$$x_0 = 1, \qquad x_1 = 2.$$

Therefore,

$$h = x_1 - x_0 = 2 - 1 = 1.$$

Also,

$$f(x_0) = 3$$

and

$$f(x_1) = 7.$$

Therefore,

$$I_T = \frac{1}{2}(3 + 7).$$

Hence,

$$I_T = \frac{10}{2}.$$

Therefore,

$$\boxed{I_T = 5}$$

5.2 Composite Trapezoidal Rule

Suppose

$$I = \int_a^b f(x)\,dx.$$

Divide the interval $[a, b]$ into n equal subintervals.

The step size is

$$\boxed{h = \frac{b-a}{n}}$$

and the points are

$$x_i = a + ih, \qquad i = 0, 1, \ldots, n.$$

Thus,

$$x_0 = a, \qquad x_n = b.$$

Applying the Trapezoidal Rule to each subinterval gives

$$\int_a^b f(x)\,dx \approx \frac{h}{2}\left[f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(x_n)\right].$$

Therefore,

$$\boxed{I_T = \frac{h}{2}\left[f(x_0) + 2\sum_{i=1}^{n-1} f(x_i) + f(x_n)\right]}$$

The coefficient pattern is

$$\boxed{1,\ 2,\ 2,\ \ldots,\ 2,\ 1}$$

Error Term

The error for the Composite Trapezoidal Rule is

$$\boxed{E_T = -\frac{(b-a)}{12} h^2 f''(\xi)}$$

for some

$$\xi \in (a, b).$$

Therefore,

$$\boxed{\int_a^b f(x)\,dx = I_T - \frac{(b-a)}{12} h^2 f''(\xi)}$$

and hence,

$$\boxed{E_T = O(h^2)}$$

This means that if the step size is reduced, the error decreases approximately according to h^2.

Example 3

Use the **Composite Trapezoidal Rule** with $n = 4$ to approximate

$$I = \int_0^2 x^2 \, dx.$$

The Composite Trapezoidal Rule is

$$\boxed{I_T = \frac{h}{2}\left[f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(x_n)\right]}$$

First, calculate the step size:

$$h = \frac{b-a}{n}.$$

Here,

$$a = 0, \qquad b = 2, \qquad n = 4.$$

Therefore,

$$h = \frac{2-0}{4}.$$

Hence,

$$\boxed{h = 0.5}$$

Therefore,

$$x_0 = 0,$$

$$x_1 = 0.5,$$

$$x_2 = 1,$$

$$x_3 = 1.5,$$

$$x_4 = 2.$$



Since

$$f(x) = x^2,$$

we obtain

$$f(0) = 0,$$

$$f(0.5) = 0.25,$$

$$f(1) = 1,$$

$$f(1.5) = 2.25,$$

and

$$f(2) = 4.$$

The table is

x	0	0.5	1	1.5	2
$f(x)=x^2$	0	0.25	1	2.25	4

$$I_T = \frac{h}{2}\left[f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + f(x_4)\right].$$

Substituting the values,

$$I_T = \frac{0.5}{2}\left[0 + 2(0.25) + 2(1) + 2(2.25) + 4\right].$$

Simplifying,

$$I_T = 0.25\left[0 + 0.5 + 2 + 4.5 + 4\right].$$

Therefore,

$$I_T = 0.25(11).$$

Hence,

$$\boxed{I_T = 2.75}$$

Calculate the Exact Value

The exact integral is

$$I = \int_0^2 x^2 \, dx.$$

Therefore,

$$I = \left[\frac{x^3}{3}\right]_0^2.$$

Thus,

$$I = \frac{8}{3}.$$

Hence,

$$\boxed{I = 2.6667}$$

Calculate the Absolute Error

$$E_a = |I - I_T|.$$

Therefore,

$$E_a = |2.6667 - 2.75|.$$

Hence,

$$\boxed{E_a \approx 0.0833}$$

Example 4

The following values of a function are given:

x	0	1	2	3	4	5
$f(x)$	2	3	5	8	10	13

Use the Composite Trapezoidal Rule to approximate

$$\int_0^5 f(x)\,dx.$$

There are 5 subintervals.

Therefore,

$$n = 5.$$

The step size is

$$h = \frac{b-a}{n}.$$

Thus,

$$h = \frac{5-0}{5} = 1.$$

The Composite Trapezoidal Rule is

$$I_T = \frac{h}{2}\left[f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + 2f(x_4) + f(x_5)\right].$$

Substitute the values:

$$I_T = \frac{1}{2}\left[2 + 2(3) + 2(5) + 2(8) + 2(10) + 13\right].$$

Therefore,

$$I_T = \frac{1}{2}\left[2 + 6 + 10 + 16 + 20 + 13\right].$$

Hence,

$$I_T = \frac{67}{2}.$$

Therefore,

$$\boxed{I_T = 33.5}$$

Thus, the approximate value of the integral is

$$\boxed{\int_0^5 f(x)\,dx \approx 33.5}$$

Difference Between Trapezoidal Rule and Composite Trapezoidal Rule

- The main difference is **how many subintervals are used to approximate the integral**.

1. Trapezoidal Rule

The ordinary Trapezoidal Rule is applied to **one subinterval**.

For

$$I = \int_a^b f(x)\,dx,$$

the entire interval $[a, b]$ is treated as one interval.

The step size is

$$h = b - a.$$

The formula is

$$\boxed{I_T = \frac{h}{2}\,[f(a) + f(b)]}$$

or, equivalently,

$$\boxed{I_T = \frac{b-a}{2}\,[f(a) + f(b)]}$$

The curve is approximated by **one straight line**, forming one trapezoid.

2. Composite Trapezoidal Rule

The Composite Trapezoidal Rule divides the interval $[a, b]$ into n smaller subintervals.

The step size is

$$\boxed{h = \frac{b-a}{n}}$$

The interval is divided as

$$a = x_0 < x_1 < x_2 < \cdots < x_n = b.$$

The Trapezoidal Rule is then applied to each subinterval.

The formula is

$$\boxed{I_T = \frac{h}{2}\left[f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(x_n)\right]}$$

The curve is therefore approximated using n trapezoids.

Simple Illustration

Suppose we want to approximate

$$\int_0^4 f(x)\,dx.$$

Trapezoidal Rule

Use the entire interval:

$$0 \longrightarrow 4$$

So there is **one trapezoid.**

$$\boxed{n = 1}$$

Composite Trapezoidal Rule

Suppose we choose

$$n = 4.$$

Then the interval is divided into

$$0, \quad 1, \quad 2, \quad 3, \quad 4.$$

Thus, we have four subintervals:

$$[0, 1], \quad [1, 2], \quad [2, 3], \quad [3, 4].$$

Therefore, there are **four trapezoids.**

$$\boxed{n = 4}$$

Example 5

Approximate

$$I = \int_0^2 x^2 \, dx$$

using:

1. The Trapezoidal Rule with one subinterval.
2. The Composite Trapezoidal Rule with $n = 4$.

Compare the results.

Part (a): Trapezoidal Rule

Using one subinterval,

$$h = 2.$$

Therefore,

$$I_T = \frac{2}{2}[f(0) + f(2)].$$

Since

$$f(0) = 0$$

and

$$f(2) = 4,$$

we obtain

$$\boxed{I_T = 4}$$

Part (b): Composite Trapezoidal Rule

For $n = 4$,

$$h = \frac{2}{4} = 0.5.$$

Using the values

$$0, \quad 0.25, \quad 1, \quad 2.25, \quad 4,$$

we obtain

$$I_T = \frac{0.5}{2}[0 + 2(0.25) + 2(1) + 2(2.25) + 4].$$

Therefore,

$$\boxed{I_T = 2.75}$$

Comparison

The exact value is

$$I = \frac{8}{3} \approx 2.6667.$$

The results are:

Method	Approximation	Absolute Error
Trapezoidal Rule	4.0000	1.3333
Composite Trapezoidal Rule	2.7500	0.0833

Therefore, increasing the number of subintervals significantly improves the approximation.

$$\boxed{\text{Smaller } h \quad \Longrightarrow \quad \text{Generally smaller error}}$$

This is an important observation to emphasize when introducing the **Composite Trapezoidal Rule.**